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I. Multiple choice questions:

1. In hot summer after a bath, the body's
   a) internal energy decreases
   b) internal energy increases
   c) heat decreases
   d) no change in internal energy and heat

2. The graph between volume and temperature in Charles' law is
   a) an ellipse b) a circle
   c)a straight line d) a parabola

3. When a cycle tyre suddenly bursts, the air inside the tyre expands. This process is
   a) isothermal b) adiabatic
   c) isobaric d) isochoric

4. An ideal gas passes from one equilibrium state (P_1, V_1, T_1, N) to another equilibrium state ($2P_1$, $3V_1$, T_2, N). Then
   a) $T_1 = T_2$ b) $T_1 = \frac{T_2}{6}$
   c) $T_1 = 6T_2$ d) $T_1 = 3T_2$

5. When a uniform rod is heated, which of the following quantity of the rod will increase
   a) mass
   b) weight
   c) center of mass
   d) moment of inertia

6. When food is cooked in a vessel by keeping the lid closed, after some time the steam pushes the lid outward. By considering the steam as a thermodynamic system, then in the cooking process
   a) $Q > 0$, $W > 0$,
   b) $Q < 0$, $W > 0$,
   c) $Q > 0$, $W < 0$,
   d) $Q < 0$, $W < 0$,

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7. When you exercise in the morning, by considering your body as thermodynamic system, which of the following is true?
   a) $\Delta U > 0$, $W > 0$,
   b) $\Delta U < 0$, $W > 0$,
   c) $\Delta U < 0$, $W < 0$,
   d) $\Delta U = 0$, $W > 0$,

8. A hot cup of coffee is kept on the table. After some time it attains a thermal equilibrium with the surroundings. By considering the air molecules in the room as a thermodynamic system, which of the following is true
   a) $\Delta U > 0$, $Q = 0$
   b) $\Delta U > 0$, $W < 0$
   c) $\Delta U > 0$, $Q > 0$
   d) $\Delta U = 0$, $Q > 0$

9. An ideal gas is taken from (P_i, V_i) to (P_f, V_f) in three different ways. Identify the process in which the work done on the gas the most.

